Machine Learning
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Adapting and Assessing Transfer Learning for Lung Disease Classification

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1. Abstract + Brief How to Summary

This report details the design, implementation, and evaluation of a deep learning-based image classification system capable of distinguishing between lung X-ray images of types Covid-19, Pneumonia and Healthy cases. Throughout was stressed, the importance of recall over accuracy, due to its position in medical diagnosis. It approached a lightweight stance on model optimality, only employing a few extra layers to an already trained model, Xception. By iteratively testing, the resulting model was deemed successful. Many metrics are used to evaluate, some of which being Grad-CAM, precision, f1 scores and a confusion matrix. The created model ended up with a macro f1-score of almost 0.99, demonstrating the importance put on the imbalanced classes and classes that are more detrimental to be false.

Database used: Lung Disease.

Dataset neatening: In a Folder called 'Lung Data' which contains the three folders, 'HEALTHY', 'COVID', 'PNEUMONIA'. All other modifications are done in the notebook.

Models trained: Xception, Transfer learning with a custom model head.

Model Review: The main folder has an 'experiments' folder, the model present in the folder called 'exp_B12_unfreezing_test' performed best, as explained in report below. All other models have been stripped of their '.keras' file, as it was too big. But the results and training logs still exist in the folder.

Model Eval: Done through detailed experimentation with a detail list on metrics evaluated and considered at each incremental stage of the experiment.

2. Introduction

Chest X-ray imaging is routinely used in clinical practise to assess & diagnose a wide range of pulmonary conditions; however, it is a task that due to extensive time and attention, the doctor can struggle to spot and classify a patient's conditions. Recent advances in ML have enabled automated analysis of chest X-ray images for the classification of pulmonary diseases. Despite this, automating the distinguishing between different pathologies (e.g. COVID, Pneumonia) remains challenging due to overlapping radiographic features and imperfect data.

In this work, a supervised classification task is formulated to distinguish between multiple lung conditions using chest X-ray images; the dataset used consists of 3 clinical categories: Covid infected, Pneumonia infected and Healthy lungs. The report highlights several challenges, including class imbalance, subtle inter-class visual overlap, increasing the model's black box visibility and the need for carefully chosen evaluation metrics beyond overall accuracy.

Throughout, we arrange the experiment to explore a plethora of different hyperparameters, optimiser choices and transfer learning strategies, to provide a successful model capable of disease classification.

3. Dataset Description + Problem Framing

This section describes the dataset used in this study and outlines the classification problem it presents.

3.1 Dataset Overview + Challenges

The dataset consists of chest X-ray images belonging to three clinically relevant categories, each exhibiting distinct characteristics and challenges. The categories being Healthy lungs, Pneumonia infected lungs and Covid infected lungs. The dataset presents several challenges for reliable classification. First, the class distribution is highly imbalanced, with significantly less Covid and Pneumonia images compared to healthy cases. Overall, the Dataset consists of 15,153, which isn't a significant amount, this could also potentially develop into an overfitting problem if the model isn't handled correctly. Second, radiographic occasions of unhealthy lungs often exhibit overlapping visual patterns, making inter-class separation

difficult. Additionally, chest X-ray images are subject to variability in disease severity and patient positioning which further increases intra-class variation. Finally, the data consists of large amounts of medical ambiguity, unlike classifications of objects like a table or a bottle that have sharp edges, the cloudiness present in a diseased lung is difficult to differentiate.

3.2 Problem Formulation

Based on the dataset composition, the task is formulated as a supervised multi-class image classification problem. Each input consists of a chest X-ray image, and the model is required to predict one of the three diagnostic categories. The objective is to learn discriminative visual representations that enable accurate classification despite class imbalances and subtle inter-class differences.

Given the medical context, performance is prioritised towards class sensitivity for disease classes over overall accuracy, to reduce the risk of false negative predictions.

4. Dataset Preprocessing Pipeline

To ensure consistency and suitability for model training, a structured preprocessing pipeline was applied to the raw image data.

4.1 Image Preprocessing

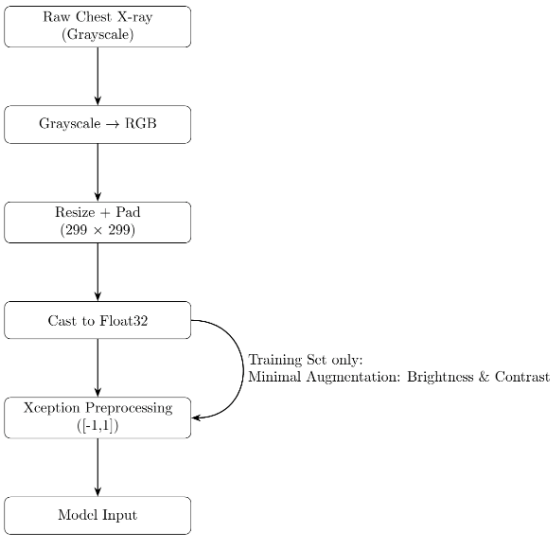

Figure 1: Preprocessing Pipeline. All samples follow a unified preprocessing path, while data augmentation is applied only during training.

Prior to training, all images underwent a series of preprocessing steps to standardise input format and resolution. Preprocessing is seen as one of the most influential steps in the whole ML Pipeline; a step that can impact the successfulness of a ML algorithm. Each decision has been thought out. The preprocessing pipeline was designed to ensure compatibility with the selected convolutional backbone, introduced in section 5. As displayed in Figure 1, all chest X-ray images were converted from grayscale to three-channel RGB to ensure compatibility with the Xception architecture pretrained on ImageNet. Images were resized to 299x299 pixels using padding to preserve aspect ratio and avoid geometric distortion of anatomical structures. The dataset was already in the 299x299 size requirement, so no size readjustments were done, despite it being in the preprocessing function. Only lightweight augmentation was intentionally introduced in order to preserve clinically relevant visual information, and was only done on the training data, leaving the validation and test set as clean and real as possible. The input was normalised in line with

the pretrained backbone to align feature distributions with pretrained weights. The preprocessing pipeline was fixed across all experiments to ensure fair and controlled comparison between training configurations.

4.2 Dataset Splitting

The dataset was partitioned into training, validation, and test subsets to enable robust model development and evaluation. The data split is specified in Table 1 below. The subsets have been broken down using a 70% ,15% and 15% split for the training, validation and test sets respectively, a validation set is included to monitor model performance during the training phase, enabling informed hyperparameter tuning and early identifications without contaminating the test set. 15 percent of the data was left out for the testing set. This dataset was used exclusively for final model performance evaluation to provide an unbiased assessment of the model's generalisation capability. The split was selected as it provides sufficient data for model training while retaining enough for validation and test sets for reliable performance estimation.

Subset	Number of Images	Percentage (%)
Training	10,606	70.0
Validation	2,274	15.0
Test	2,273	15.0
Total	15,153	100.0

Table 1: Dataset frequency split across Training, Validation and Test set

The split was performed in a stratified manner, ensuring that class proportions were preserved across all subsets. Class labels were encoded as follows:

0 = Healthy, 1 = Pneumonia, and 2 = COVID.

The Batch size got set to 16 as it is large enough to fit into the GPU memory with an image size of 299x299, but also not too small where training slows considerably.

4.3 Class Imbalance Handling

Given the uneven distribution of classes within the dataset, specific measures were taken to mitigate the effects of class imbalance. Class imbalance can be dangerous for model training as it can bias certain majority classes, skewing classification direction. In this case, favouring the majority 'Healthy' class can potentially provide dangerous classification for deployment.

To mitigate the effects of class imbalance, class weights were computed from the training data and incorporated into the loss function during training. This ensured that majority classes contributed proportionally to optimisation while preserving the natural distribution of samples in the validation and test sets.

Other options were assessed. Undersampling loses a significant amount of important data in the majority class. With a dataset of this size, removing data samples could hurt training with a threat of underfitting. Oversampling generates new samples for the underrepresented classes. Synthetically generating medical images pose a large risk of the model learning false/artificial special information. Oversampling can also be done by repeating true minority images, balancing the split; despite not having the issue mentioned previously, it can have an increased chance of overfitting.

Class weighting was deemed more logical as it doesn't introduce synthetic bias into clinically sensitive classes and it doesn't remove important data.

5. Model Architecture

This section details the neural network architecture employed for lung disease classification.

5.1 Architectural Context and Model Selection

Over the last decade, many solutions have arisen for the problem of image classification, the main of which being Convolutional Neural Networks (CNN). CNN's employ the concept of learnable filters which can scan over an image matrix extracting features and patterns to derive a classification.

Each CNN can vary in layer types, quantities and arrangement of layers and other tuneable parameters to create a dynamic and unique solution to a problem at hand. Given the dominance of convolutional architectures in image classification, several established CNN backbones were considered for this task. The table below shows a few of the highest quality, trusted models that were ideologically evaluated and compared for this project [1].

Model	Size	Parameters	Depth
Xception	88 MB	22,910,480	126
VGG19	549 MB	143,667,240	26
ResNet50	98 MB	25,636,712	-
ResNet101	171 MB	44,707,176	-
InceptionV3	92 MB	23,851,784	159

Table 2: Various CNN Models being compared on size

The Xception architecture was adopted as our convolutional backbone, given that the radiographic images are provided at a native resolution of 299×299. Xception's depth wise separable convolutions are well suited to capturing fine-grained texture patterns common in chest X-ray imagery, while its native input resolution avoids unnecessary rescaling and preserves diagnostic detail. Despite its size, both in depth and parameter count, being not excessively small, comparatively against the bigger models like VGG19 and ResNet101, it is much more size efficient therefore being a large factor against the larger models. This choice provides a strong and computationally efficient baseline for subsequent experimental evaluation.

5.2 Architecture Design

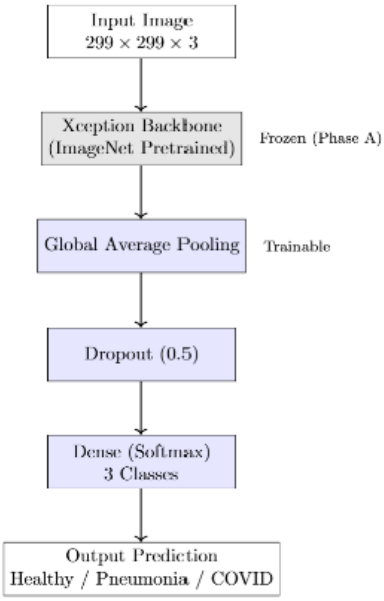

Figure 2: Transfer learning architecture based on an ImageNet-pretrained Xception backbone. During baseline training, the convolutional backbone is frozen while the task-specific classification head is

trained. In the fine-tuning phase, higher convolutional layers are selectively unfrozen to enable domain adaptation.

A transfer learning approach was adopted to leverage prior knowledge from large-scale image datasets, therefore, the final model architecture combines a pretrained convolutional backbone with a lightweight task-specific classification head. Using this approach, we dramatically increase training speed and model successfulness; in a project limited by GPU usage, this is hugely beneficial. As mentioned previously, our backbone stems from the weights of an ImageNet pretrained Xception. The overall model architecture and transfer learning strategy are illustrated in Figure 2.

One might notice that there are few additional layers, the task specific classification head was intentionally kept lightweight to reduce overfitting and preserve the representational power of the pretrained backbone. Medical datasets are prone to overfitting due to large class imbalances and subtle feature differences, which this dataset also consists of, hence, the decision to keep the simple head; the role of the backbone is to keep the deep feature extraction that Xception can perform well and have the head's role be for mapping custom classifications and less for learning vision skills.

Only three layers, consisting of a Global Average Pooling Layer, a Dropout Layer and a single Dense Layer, were introduced to the model architecture, each chosen for their own reasons. Global Average Pooling was employed to aggregate spatial features while avoiding the parameter explosion associated with fully connected flattening layers; with this layer, it can reduce spatial dimensions without introducing extra parameters, force the model to learn more global and class relevant features and therefore, improves robustness for chest X-ray images where lesions are subtle and spatially diffuse. This is preferable to Global Maximum Pooling in this case as that risks discarding weak but clinically relevant signals in favour of isolated high activation regions, Average Pooling identifies distributed textures like those in a pathology. The next step of this custom neural network is a Dropout Layer, applied to act as a regularisation step, randomly switching off some of a layer's neuron. It solves a few different problems, one being the issue of co-adaptation, the idea that multiple neurons extract similar feature information through nearly identical connection weights. This is a problem as it can lead to overfitting and a wastage of machine resources. Then a single fully connected layer was used to map learned representations to class probabilities, as a deeper head were unnecessary given the expressive capacity of the backbone network.

Overall, deeper classification heads were deliberately avoided, as preliminary analysis indicated an increased risk of overfitting without corresponding gains in generalisation. This architectural design prioritises generalisation, stability and interpretability while avoiding unnecessary parameter growth in the classification head.

Additionally, a focus was on adapting hyperparameter, optimiser choice and other transfer learning strategy variables over the increase of randomly put together additional layers that increase complexity and model efficiency. Thus, this stayed constant throughout the rest of the experimentation.

6. Experiment Design

A series of controlled experiments was designed to systematically evaluate the impact of different training configurations, and each experiment was conducted under controlled conditions to allow fair comparison between model variants.

6.1 Unified Training Protocol

A moderate epoch budget was used across all experiments, with early stopping employed to prevent unnecessary training once convergence was achieved. Preliminary exploratory runs indicated that convergence typically happens at around 5-12 epochs, 15 was chosen to give enough headroom avoiding risk of underfitting and allows `EarlyStopping` to decide. Early stopping with a patience of 5 epochs was chosen to halt training once validation performance ceased to improve, reducing the overfitting risk; Anything smaller was seen too aggressive and anything larger saw too much wasted time.

To support stable convergence and reduce near-minima oscillations, `ReduceLRonPlateau` was employed which analyses progress every 3 epochs then evaluates if the learning rate needs to be changed.

Across all experiments, the following factors were kept constant:

- Dataset split
- Preprocessing pipeline
- Base architecture (Xception)
- Class weighting strategy
- Evaluation metrics
- Early stopping criteria

6.2 Baseline Configuration

The baseline model employed a frozen Xception backbone with only the classification head trained. As this phase involved training newly initialised layers, a relatively high learning rate was selected to allow rapid adaptation and efficient optimisation of the classifier head.

Stochastic Gradient Descent with momentum was considered; however, an adaptive optimisation strategy was selected as more appropriate for this task. This phase sees only the newly initialised layers trained, requiring rapid and stable convergence from random initialisation. The Adam optimiser was therefore chosen due to its adaptive learning rate mechanism and its first and second moment estimates, which enable faster convergence and reduce sensitivity to learning rate selection compared to SGD. Given the limited dataset size, class imbalance and subtle inter-class differences inherent in medical imaging, Adam provided a more robust optimisation strategy while reducing the need for extensive hyperparameter tuning. Overall, SGD with Momentum is often more effective once feature representations are well-aligned with the target domain whereas this work prioritised stable optimisation during transfer learning.

6.3 Controlled Experiment

Model improvements were introduced progressively along 3 axes: training hyperparameters, optimiser choice and transfer learning strategy. Each axis was explored independently to isolate its effect on performance and avoid confounding.

Only one experimental factor was varied at a time.

Rather than performing sequential finetuning within a single training run, each finetuning configuration was trained independently with a fixed freezing strategy. This ensured fair comparison and avoided confounding effects arising from different optimisation histories; observed performance differences were attributable to the configuration itself rather than inherited optimisation dynamics such as optimiser momentum, learning rate schedules or accumulated gradient history. As a result, conclusions drawn from the experiments can be causally linked to the design choices under investigation.

The full set of experimental configurations derived from this controlled design is summarised in Table 3. The table presents the exploration of training hyperparameters, optimiser choice and unfreezing strategy,

with each experiment differing by a single controlled variable. This structured layout ensures transparency, reproducibility and clear traceability between intent and observations.

Stages 1-3 were only trained for 4 epochs as it was only for head specialisation.

For Stage 4, we introduced the idea of unfreezing and training a specified frequency of already trained weights within the Xception base model. This is the true transfer learning. Training was performed in two phases. In the first phase, the pretrained backbone was frozen and only the classification head was trained using a learning rate + weight decay gathered from previous experiments to enable the rapid adaptations. Training the head for 4 epochs. In the second phase, selected backbone layers were unfrozen, and the entire model was fine-tuned using a significantly smaller learning rate to preserve pretrained representations while adapting the model further to the domain required. Training of this phase was set at 15. Leading to a total of 19 epochs for these models.

6.4 Experiment Configuration

Stage 0 — Reference Baseline

Exp ID	Backbone	Frozen	Optimiser	LR	WD	Notes
B0	Xception	Yes	Adam	1×10^{-3}	0	Phase A baseline

Stage 1 — Learning Rate Sensitivity (Frozen Backbone)

Exp ID	Optimiser	LR	WD	Expected Insight
B1	Adam	1×10^{-3}	0	Fast convergence
B2	Adam	3×10^{-4}	0	Stability trade-off
B3	Adam	1×10^{-4}	0	Conservative baseline
B4	Adam	3×10^{-5}	0	Risk of underfitting

Stage 2 — Weight Decay Regularisation

Exp ID	Optimiser	LR	WD	Purpose
B5	Adam	Best	1×10^{-4}	Reduce overconfidence
B6	Adam	Best	1×10^{-5}	Mild regularisation

Stage 3 — Optimiser Comparison

Exp ID	Optimiser	Momentum	LR	WD	Rationale
B7	Adam	–	Best	Best	Adaptive baseline
B8	AdamW	–	Best	Best	Decoupled regularisation
B9	SGD	0.9	Best	Best	Generalisation test

Stage 4 — Unfreezing Strategy

Exp ID	Frozen Layers	Backbone LR	Head LR	Strategy
B10	All	–	–	Frozen control
B11	Top 20%	1×10^{-4}	Best	Partial fine-tune
B12	Top 40%	1×10^{-4}	Best	Deeper adaptation
B13	None	1×10^{-4}	Best	Full fine-tune

Table 3: Staged experimental design outlining the progressive optimisation of training hyperparameters, optimiser choice, and transfer learning strategy for the Xception-based classifier.

7. Evaluation Metrics

Model performance was assessed using multiple evaluation metrics to capture different aspects of classification behaviour.

7.1 Limitations of Accuracy

Overall classification accuracy alone is insufficient for assessing performance on imbalanced medical datasets. A model can achieve high accuracy favouring the 'Healthy' class, the majority class, hiding poor performance on minority disease classes. Accuracy assigns equal weight to all misclassifications despite the significantly different clinical implications of false negatives and false positives. From a clinical perspective, false negatives, particularly failing to identify diseased lungs is critical for the patient.

Although, recall based metrics where prioritised during evaluation, accuracy was retained as the training metric due to implementation constraints encountered when computing per-class recall during batch wise training.

7.2 Metrics Used

To address these limitations, a range of complementary metrics was employed. These are evident in Table 4 below.

Metric	**Level**	**Purpose**	**Primary Use**
Accuracy	Global	Measures the proportion of correctly classified samples across all classes. Simple and interpretable but can be misleading in imbalanced medical datasets where majority classes dominate performance estimates.	Training Monitoring
Recall (Per-Class)	Class-wise	Quantifies the ability to identify samples belonging to each class. Recall is critical for disease classes, as low recall corresponds to missed diagnoses (false negatives).	Primary Comparison
Precision (Per-Class)	Class-wise	Quantifies the proportion of predicted positives that are correct. Precision helps analyse false positive behaviour, particularly important to avoid unnecessary clinical interventions.	Error Analysis
F1 Score (Per-Class)	Class-wise	Provides a mean of precision and recall, balancing sensitivity and prediction reliability. This metric is useful when both false positives and false negatives are undesirable.	Supporting comparison
Macro F1 Score	Global	Aggregates F1 scores equally across all classes, preventing dominant classes from masking poor minority-class performance. This provides a class-balanced summary of overall model effectiveness.	Model Ranking
Normalised Confusion Matrix	Global	Visualises class-wise prediction distributions and common misclassification patterns. Normalisation enables fair comparison across classes with unequal sample sizes.	Diagnostic Analysis
Precision-Recall Curve	Class-wise	Illustrates the trade-off between precision and recall across classification thresholds.	Threshold Behaviour

		Particularly informative under class imbalance, where ROC curves may present an overly optimistic view.	
ROC Curve	Class-wise	Used as a supplementary metric for completeness rather than primary decision-making.	Secondary analysis
Calibration Curve	Global	Assesses how well predicted probabilities reflect true outcome likelihoods.	Confidence Reliability
Grad-CAM	Spatial	Provides visual explanations by highlighting image regions contributing most to model predictions. Used to assess whether the model focuses on clinically relevant lung regions rather than spurious artefacts.	Interpretability

Table 4: Summary of evaluation metrics used to assess model performance, reliability, and interpretability under class imbalance and clinical constraints.

Our Primary Goal is Recall, however, use a multitude of other supplementary metrics are used to make a more informed decision.

7.3 GRAD-CAM

In addition to quantitative metrics, model interpretability was assessed using Gradient-weighted Class Activation Mapping (Grad-CAM)[3]. Looking to evaluate and improve a model is a difficult task; this is due to the black box nature of a neural network. However, we can improve this visibility issue by implementing the Grad-CAM algorithm.

This algorithm utilises the idea that individual layers hold important information in their trained feature maps and from this we can extract important regions that the layer focused on to produce a heatmap highlighting what part of the image the model focused on influencing its decision. When overlapping the heatmap on top of the image, focused regions will highlight, with this we can make assumptions on what the model liked to see to differentiate classes and further, can use this to make adaptation choices. Figure 3 provides a great visualisation of what was identified in the chest X-Ray using the Grad-CAM algorithm, to make its detection predictions. This example shows a certain problem area on the left of the lungs image.

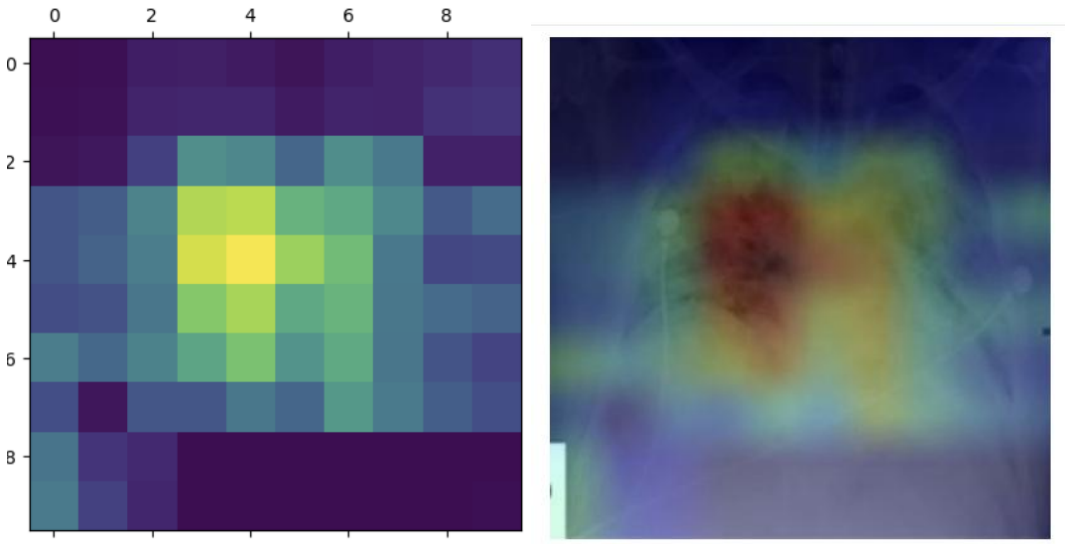

Figure 3: Two images active learning zones highlighted from the Grad-CAM algorithm. Showing what our best model does in one of its layers focuses on to make its predictions.

To implement, a tutorial was followed[2].

8. Results

This section presents the results obtained from the experimental evaluation of the proposed models.

8.1 Quantitative Results

Quantitative performance metrics are reported to compare classification performance across different experiments.

As each training step was done in a formulaic way with clear saving logs, comparing quantitative performance is justifiable and simple. In each stage of the experiment, we assess the best performer for it to progress to the next stage, as shown in Table 3. For the first stage, we evaluated the importance of Learning Rate on the training of the head classifiers; as suspected from prior research, a comparatively larger learning rate was more preferrable. As Learning rate decreased, so did the results. This is evident in Table 5 below. We require quick changes when training just the untrained classifier head, hence, its effectiveness.

By using Macro F1 to judge we passed the value of '1e-3' or '0.001' for the head LR for the remainder of the tests.

The Second stage saw no benefit from changing the weight decay. Weight decay is seen as negligible due to it being better for marginal gains when model overfits. For training a model's head, this obviously wasn't very effective at not needed. Therefore, all rows were evaluated, and Weight Decay was taken for '0.0'.

The last test for the models' head was judging optimiser choice. SGD included Momentum at 0.9 and a variant of adamw were tested, however, it didn't seem to provide any benefit. Yet again, initial assumptions made at the start of the project cycle, prevailed, hence, why our baseline seems to be the best in terms of our performance metrics.

Exp ID	LR	Weight Decay	Optimiser	Epochs	healthy recall	pneumonia recall	covid recall	macro f1
B0	1×10^{-3}	0	Adam	4	**0.938522**	0.970297	0.833948	**0.899390**
B1	1×10^{-3}	0	Adam	4	0.871158	0.955446	0.905904	0.879649
B2	3×10^{-4}	0	Adam	4	0.812296	0.980198	0.883764	0.821878
B3	1×10^{-4}	0	Adam	4	0.756050	0.980198	0.850554	0.779711
B4	3×10^{-5}	0	Adam	4	0.711576	0.970297	0.769373	0.722052
B5	1×10^{-3}	0.0001	Adam	4	0.824722	0.980198	**0.926199**	0.854947
B6	1×10^{-3}	0.00001	Adam	4	0.916285	0.985149	0.848708	0.883676
B7	1×10^{-3}	0	Adam	4	0.829954	**0.995050**	0.922509	0.856916
B8	1×10^{-3}	0	AdamW	4	0.898627	0.950495	0.891144	0.892213
B9	1×10^{-3}	0	SGD	4	0.810334	0.945545	0.891144	0.827070

Table 5: Summary of experimental configurations for model head training evaluated in this study.

After we found the conditions for optimal head training, we went on the full finetuning. Here we trained a head and then went one step further, unfreezing the top 0,20,40,100 % respectively, in a second training step. The Results saw a great increase in recall in all accounts. These extra trainable parameters, allowing for slight tweaking of pretrained weights, saw the recall go up to 0.98/0.99 consistently. A staggering update. That being said, by looking at Table 7, you can see the large difference of the number of trainable parameters between the top 20% and completely frozen.

Exp ID	Head LR	Finetuning LR	Weight Decay	Optimiser	Head Epochs	Finetuning Epochs	healthy recall	pneumonia recall	covid recall	macro f1
B10	1×10^{-3}	1×10^{-4}	0	Adam	4	15	0.896664	0.980198	0.892989	0.88908
B11	1×10^{-3}	1×10^{-4}	0	Adam	4	15	0.988882	**0.985149**	0.987085	0.981619
B12	1×10^{-3}	1×10^{-4}	0	Adam	4	15	0.991498	**0.985149**	**0.992620**	**0.986877**
B13	1×10^{-3}	1×10^{-4}	0	Adam	4	15	**0.993460**	**0.985149**	0.988930	0.986494

Table 6: Summary of experimental configurations for full model finetune training evaluated in this study. Each of these models have exposed a different number of parameters to be trianable

All Things considered B12, was decided as the best due to its higher macro f1 than all other trials. Additionally, it seems to do better on minority recalls, which was our main desire, hence, the choice for the best model.

Exp ID	Top Percentage	Trainable Parameters	No. of Trainable Layers
B10	0	6,147	3
B11	20	8,408,507	30
B12	40	12,712,443	56
B13	100	20,867,627	135

Table 7: Summary showing the extenet to how much of the model get trained in each of the finetuning efforts

8.2 Graphical Analysis

Graphical analyses are used to further illustrate model behaviour and performance trends. All graphs precent are from Model B12, training with top 40% unfrozen.

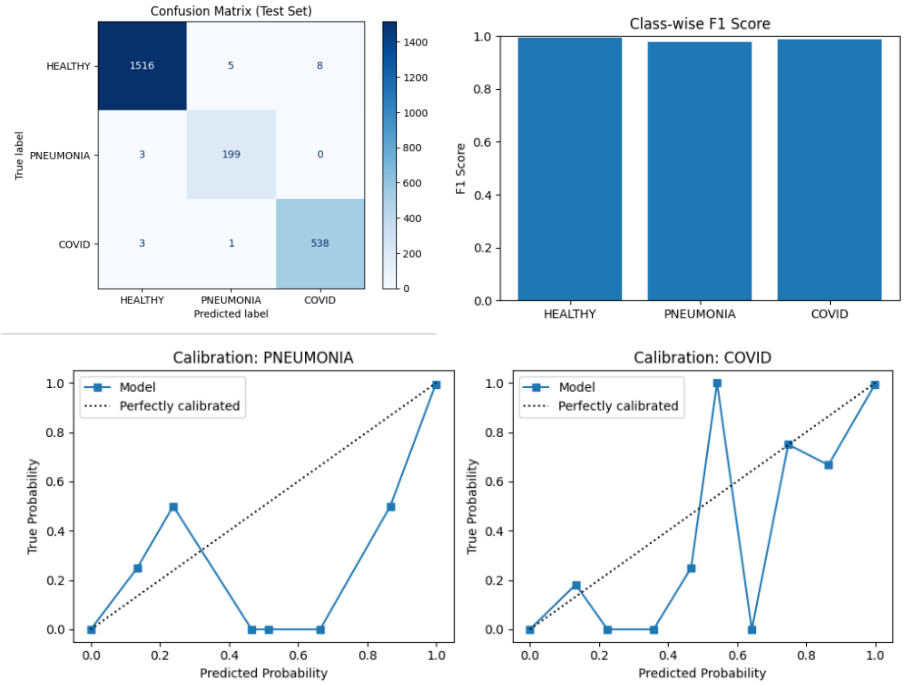

Figure 4: Graphical representation of results. Confusion Matrix (top left), Class wise f1 score (top right), Calibration curve for diseased cases (bottoms)

9. Discussion

	precision	recall	f1-score	support
HEALTHY	0.996058	0.991498	0.993773	1529.0
PNEUMONIA	0.970732	0.985149	0.977887	202.0
COVID	0.985348	0.992620	0.988971	542.0

Table 8: Results table for various classes under the testing dataset with the best model from the experiments

The Final Quantitative result to focus on is Table 8. This table shows the overall success of this trained model, on counts of precision, recall and f1 score for each class. It doesn't seem to have a blaringly obvious discrepancy of class distribution success, our initial worry, it can confidently be said that class weighting was a successful solution to class imbalance. We see a slight overconfidence issue in the diseased classes within the Calibration curve seen in Figure 4, this could still signify the underlying problems of class imbalances. However, when looking at the true testing set, the model was very successful, so it could be due to the large depth of the Xception model.

Transfer learning has solidified itself as an extremely viable option for domestic ml engineering. During this project, I had a few issues with memory overload from pushing too high of a batch size; training on a free account of google colab is extremely difficult but having the ability to profit off the parameter weights of an already established model is helpful. In only the 4 epochs added during the first stage of transfer learning, the model could gather an accuracy of between 70 to just under 90 percent, in certain runs. An exceedingly quick and relatively satisfactory result. Logically, it wouldn't be clinically ready, so it isn't perfect, but such a large increase in small amount of training epochs is a very attractive attribute of this training cycle. For a further 15 epochs, it was able to get to a stage of only a small handful of wrongdoings, which is very near to being perfect, the obvious goal of any diagnosis ml algorithm. This is extremely promising for the whole medical field, as it can bring a considerable amount of global altruistic utility.

10. Limitations + Future Work

Despite promising results, several limitations of the current approach remain. The dataset is still on the smaller side, therefore, this can benefit from a larger network of images. This project purposefully kept the amount of augmentation, a typical data scarcity aid, to a minimal due to its potential to disrupt training on medical images as they contain so much small intra-class detail that needs to be detected. However, even with this, solutions are arriving and it can still be attempted as a use to buff a dataset and increase its ability to handle variance. This would be interesting to see if this would improve or hinder the model.

This project took the active decision to not include a huge model head when customising the Xception model. This came through a few tests and personal intuition, seeing that it doesn't need to be super complex for great results, and this project has proven that. There definitely could be a layer combination out there that would largely benefit this use case. It would be interesting to find this solution.

Additionally, this project employed a sequential grid search type hyperparameter search, it would have been amazing to include a more adaptive and dynamic search, such as Bayesian hyperparameter optimisation search, which uses statistical relationships to find the next combination.

11. Conclusion

This project investigated the use of transfer learning for multi-class lung disease classification from chest X-ray images, using a lightweight model head. It had a clear plan to experiment incrementally and find a solution that perfectly executes the machine learning pipeline.

1. References

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